

# Design and Experimental Evaluation of a ROS-Based Social Robot for Multimodal Mood and Health Monitoring in Office Environments

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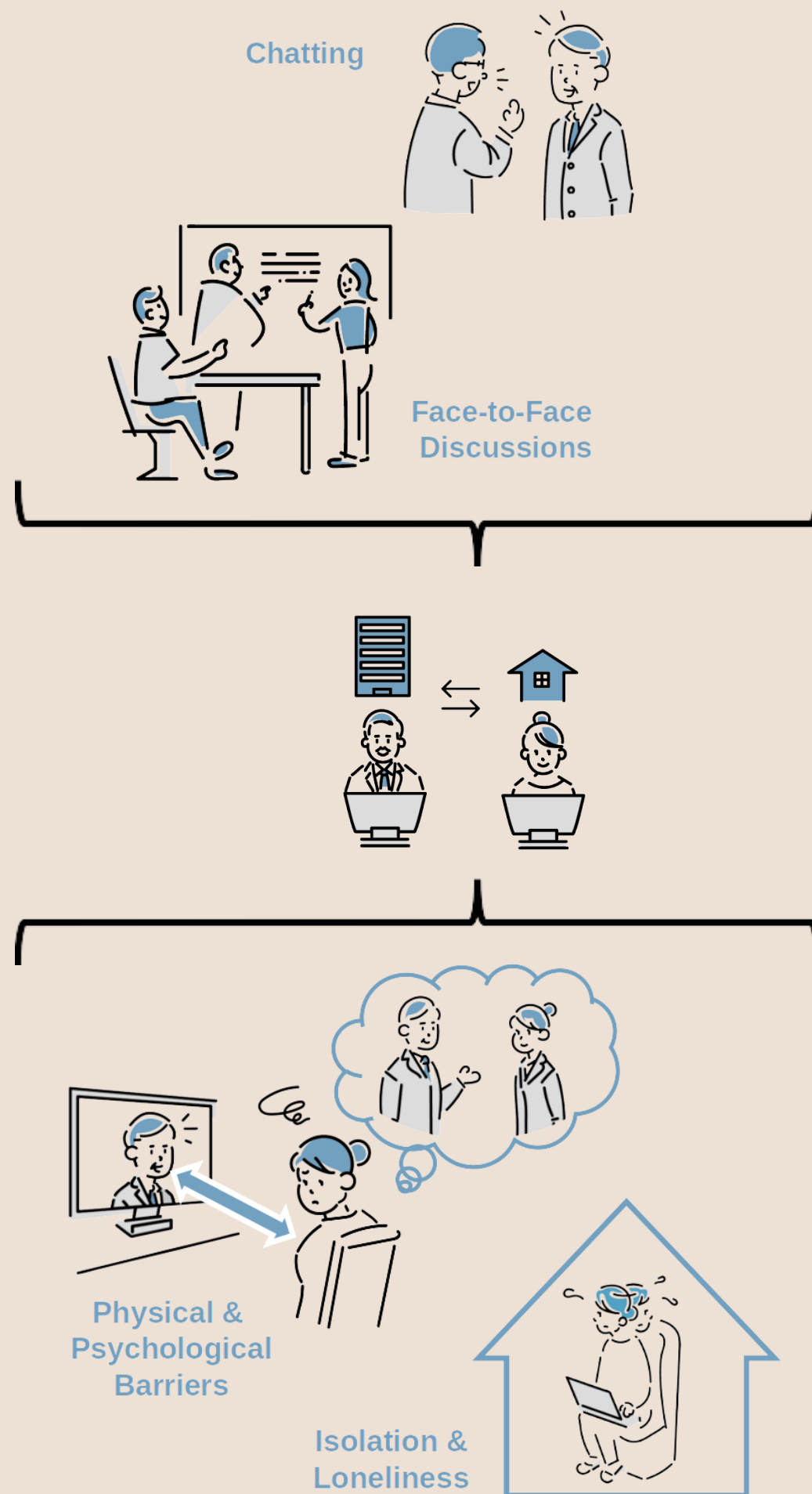
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## • Context.

Rising stress and mental-health concerns among young professionals in office settings; organizations need unobtrusive ways to support well-being.

## • Problem.

Existing systems often rely on **single-modality** signals, **lab-only** setups, or **invasive** sensors (e.g., wearables/EEG), and rarely adapt to cultural/workplace norms.

## • Opportunity

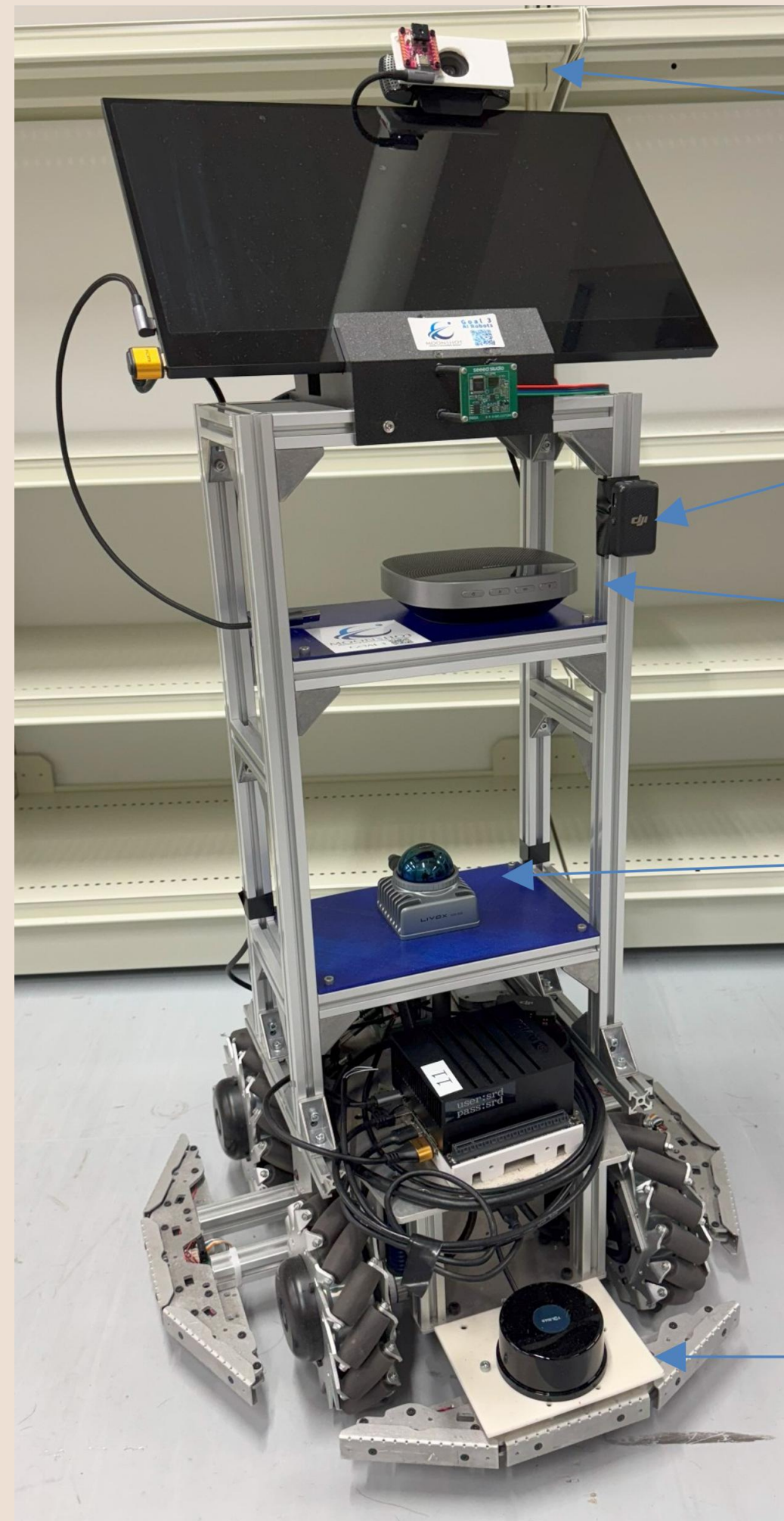
A **non-invasive, privacy-preserving social robot** that estimates mood and basic health cues and delivers supportive interactions during daily work.

## • Research Questions:

- **RQ1:** Can non-invasive multimodal sensing (FER, SER) reliably estimate employees' mood/health in real time?
- **RQ2:** Do **data-adaptive** robot interventions measurably improve user experience/mood?



- **Base:** Vstone mecatum mobile platform (v3.0).
- **Compute:** NVIDIA Jetson AGX Xavier (CUDA accelerated).
- **Cameras:** Logitech C920 (RGB); FLIR Lepton 3.5 (thermal via PureThermal3).
- **Radar:** 60 GHz mmWave (Seeed R60A) for heart rate/activity.
- **Audio I/O:** Anker PowerConf S500 (speaker), DJI Mic (demo in noisy setups).
- **LiDAR:** YDLIDAR TG30 (2D, low height) + Livox Mid-360 (3D, center).
- **Display:** 1920×1080 touch LCD (not used in tests; for future use).



Web  
camera

microphone

speaker

3DLiDAR

2DLiDAR

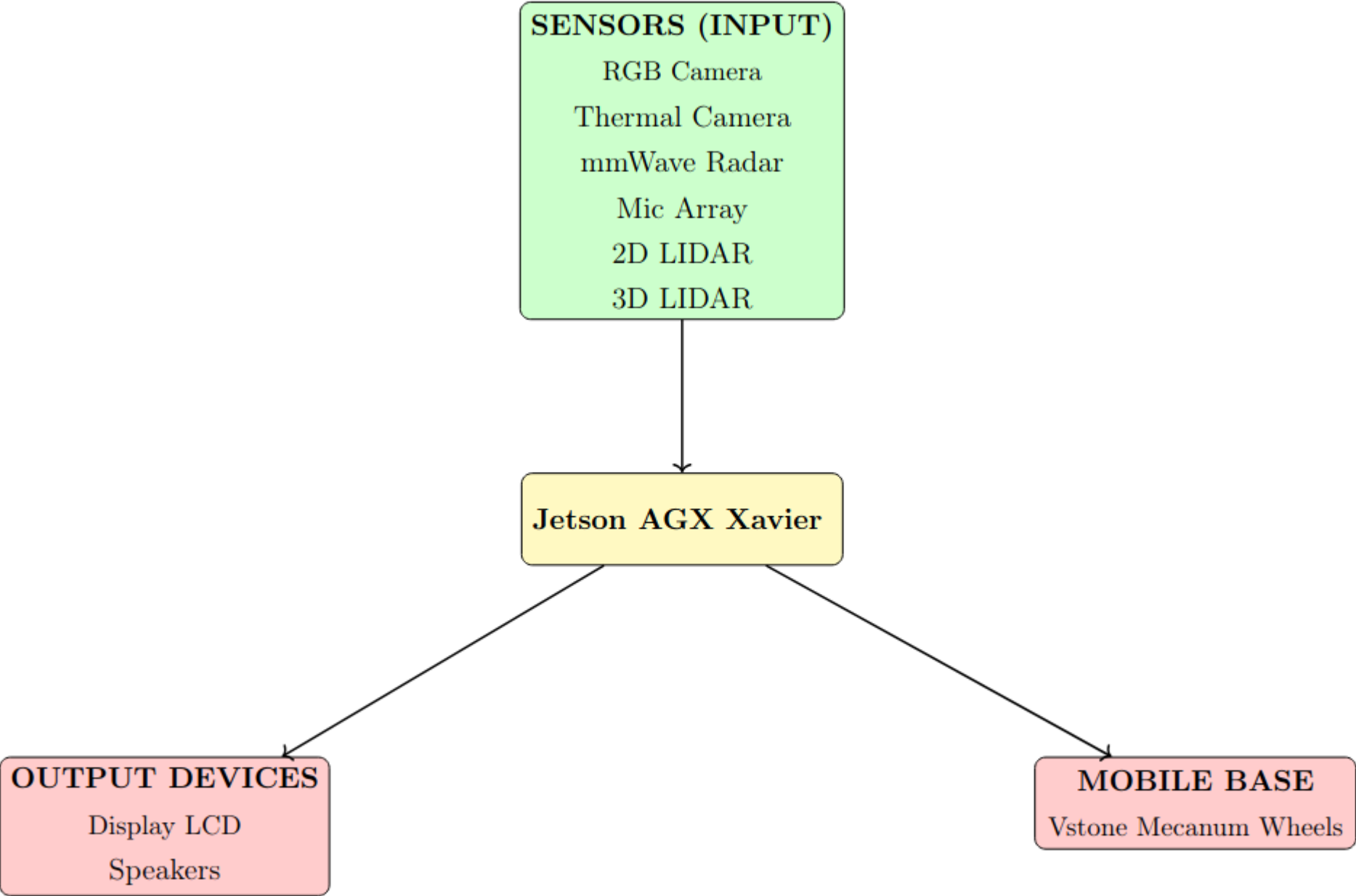
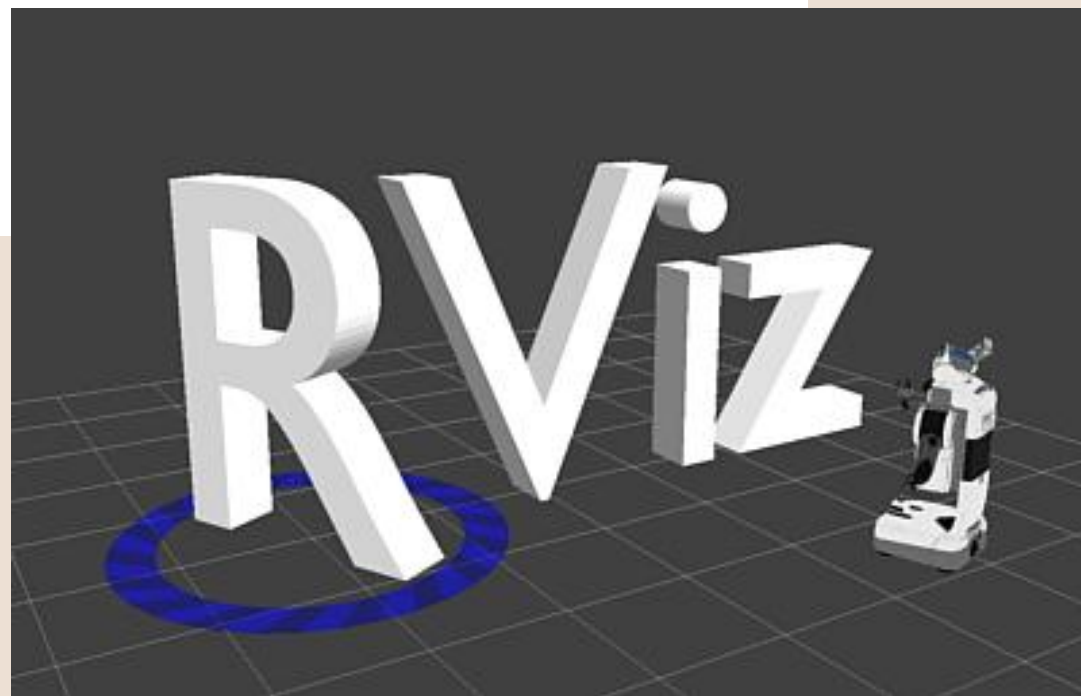


Figure 4.2: Hardware block diagram of the SUNNY robot.

The ROS logo, consisting of a 3x3 grid of blue dots followed by the letters "ROS" in a bold, dark blue, sans-serif font.

## Architecture

- **ROS Noetic** with modular nodes:  
FER, SER, thermal, mmWave, fusion, markers, navigation, interaction.
- **Topics:**  
/people\_data, /people\_data\_final (fused),  
/ser\_results, /thermal\_img, /heartrate, /LiDAR scans,  
/move\_base.
- **Logging:** per-user CSV + daily CSV for offline analysis.
- **Visualization:** RViz
- **Layers:**
  - Perception
  - Navigation
  - Estimation
  - Interaction



# Facial Expression Recognition (FER)

- **Detection:**

YuNet — fast, lightweight face detector.

- **Recognition (ID):**

SFace — embeddings; matched against **opt-in** lab photo set (filename encodes name & preferred language).

- **Expression:**

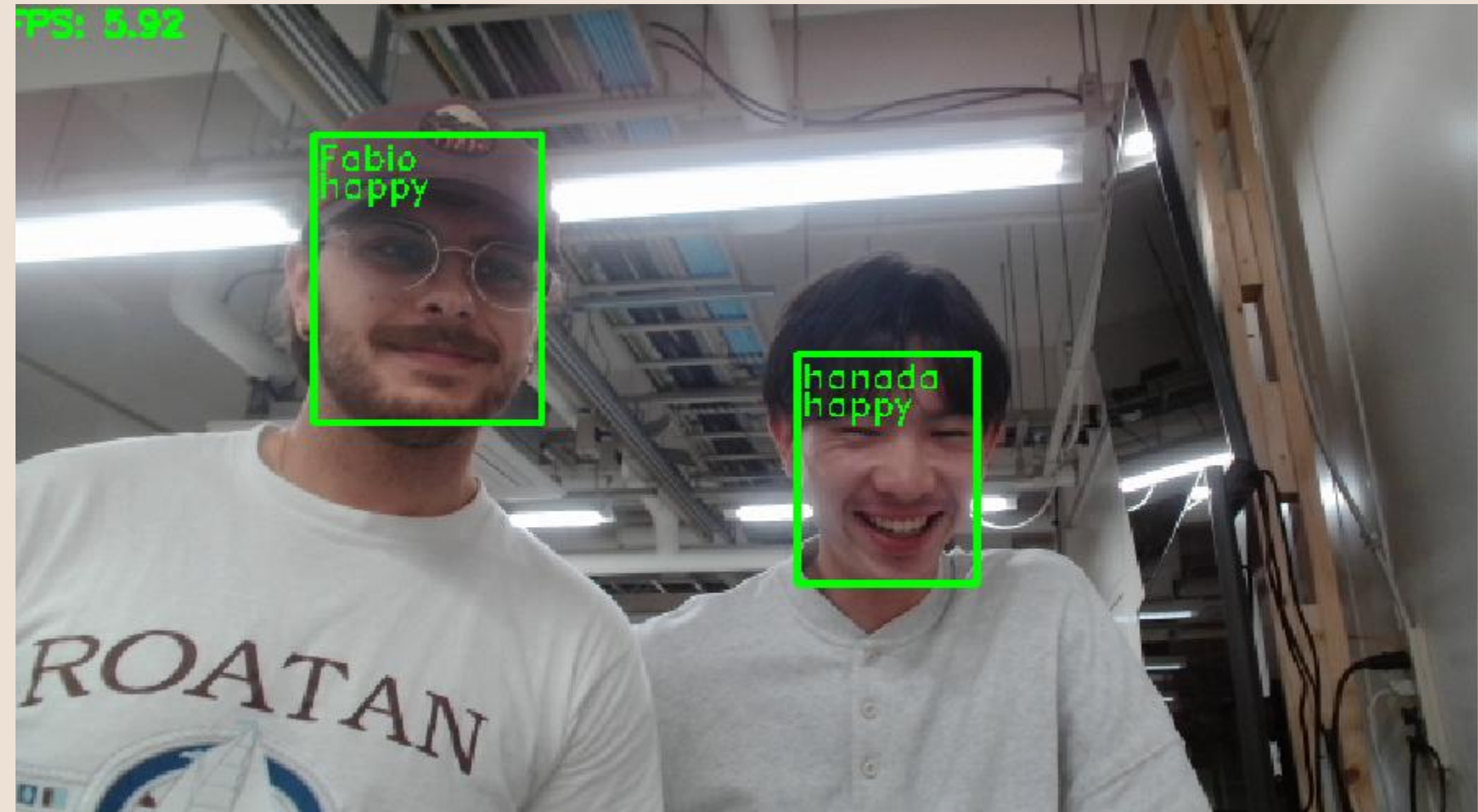
MobileFaceNet-based model for emotion classification.

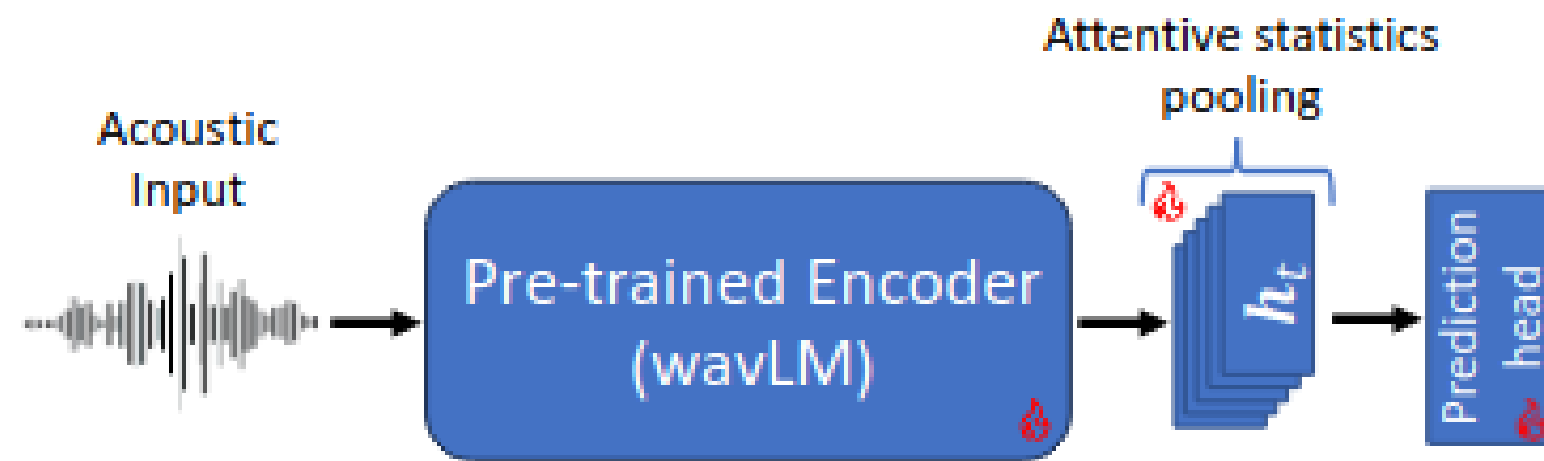
- **Runtime:**

C++/OpenCV DNN on Jetson (CUDA).

- **Privacy:**

No biometric DB storage; only features and labels in fusion/logs.





## Speech Emotion Recognition (SER)

- **Model:**

SER-Odyssey Baseline — WavLM  
AudioClassifier (categorical).

- **Labels:**

Unified and adapted to **FER** (Angry, Sad, Happy, Surprise, Fear, Disgust, Contempt, Neutral).

- **Preprocess:**

librosa load @ model SR; mean/std normalization; mask.

- **Output:**

Top-1 categorical emotion (topic / ser\_results).

- **Dataset context:**

MSP-Podcast (large, spontaneous speech).





# Thermal & mmWave

- **Thermal:**

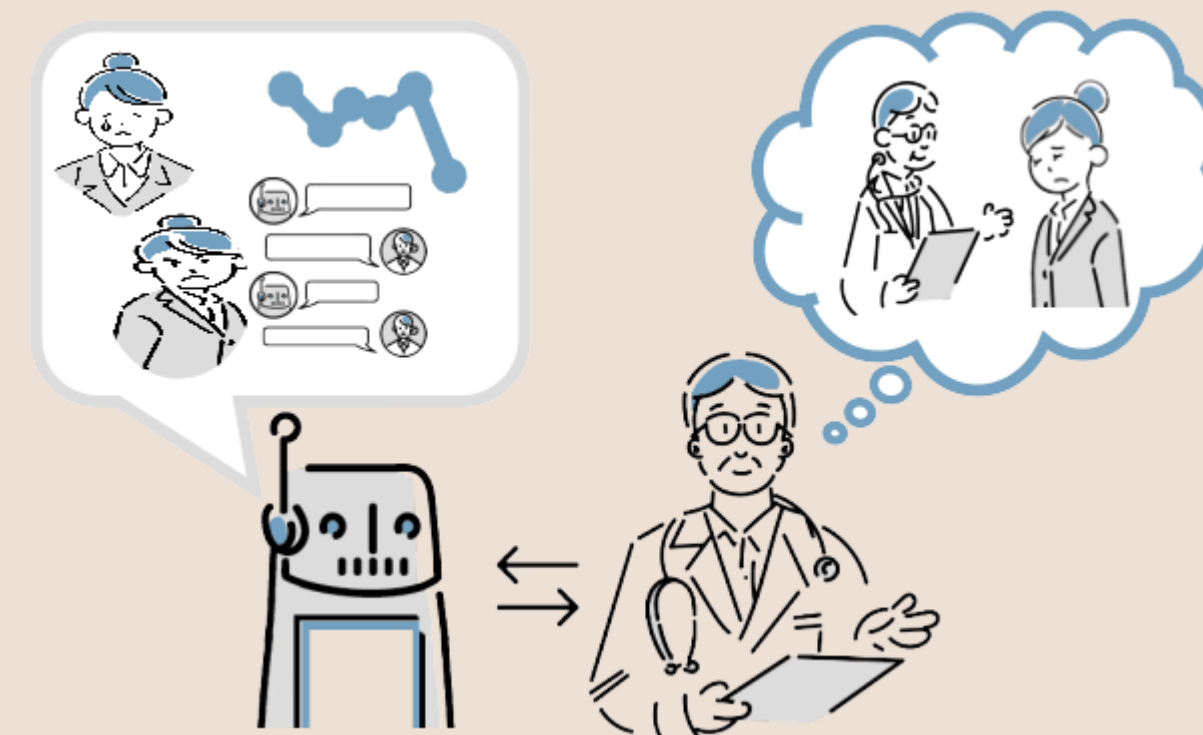
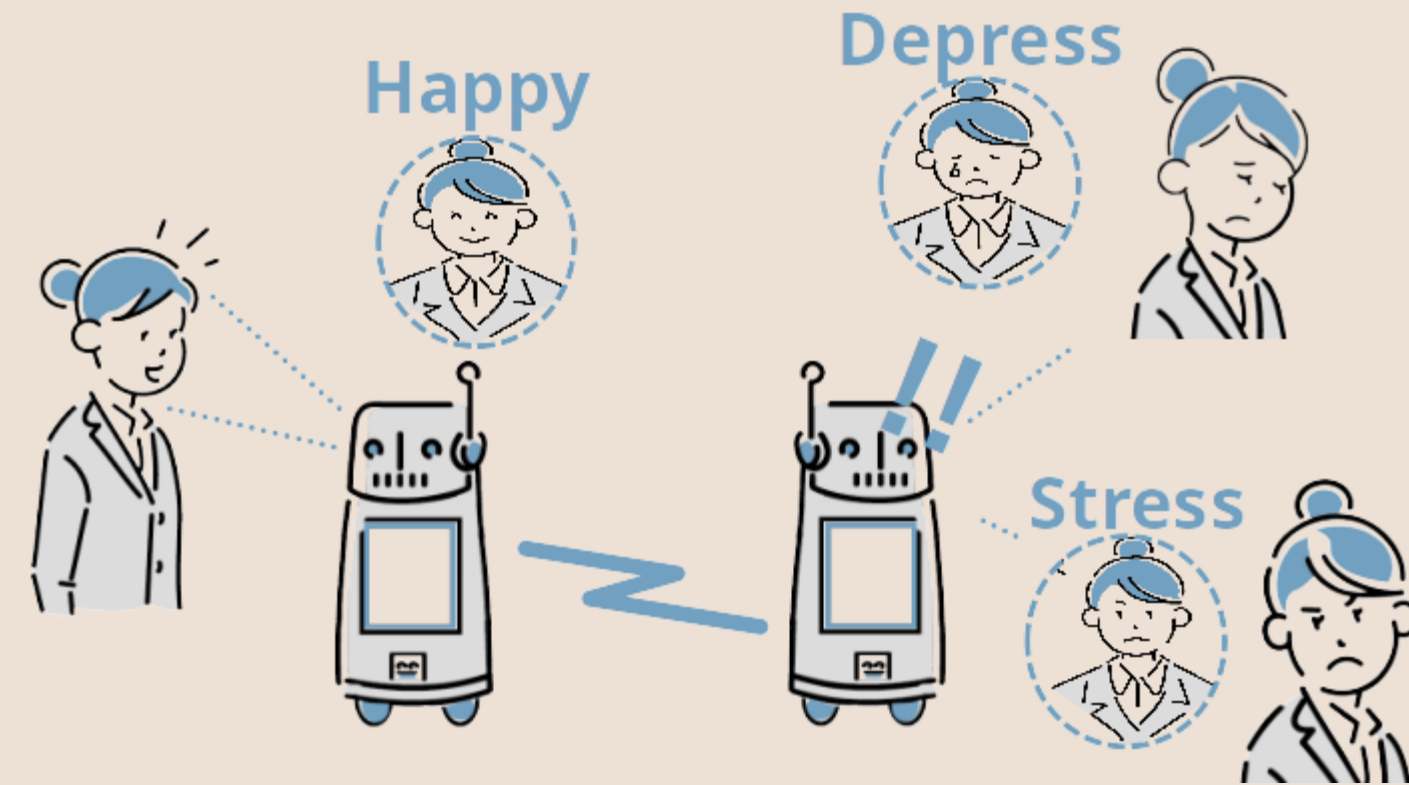
FLIR Lepton 3.5 — **fever screening** only  
(forehead estimate), no raw frames stored.

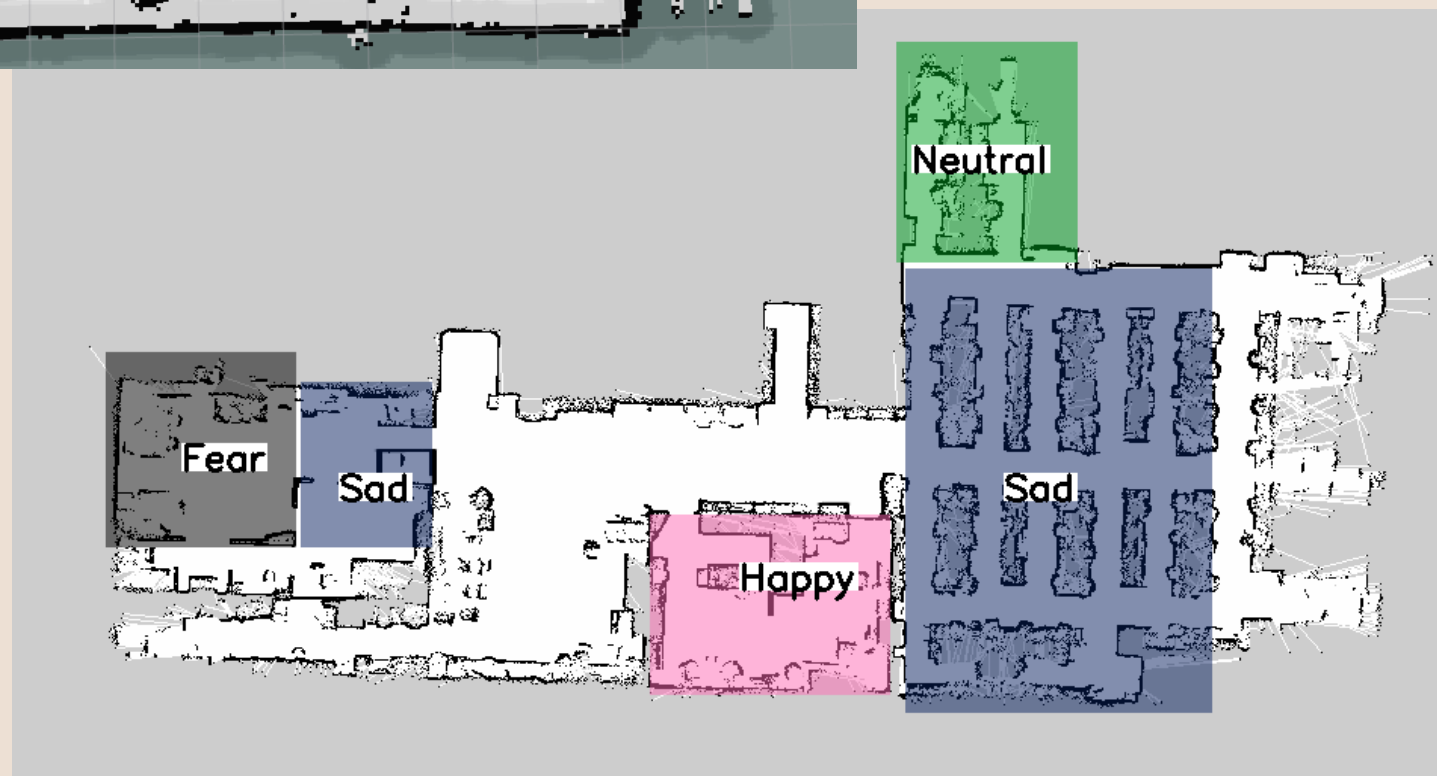
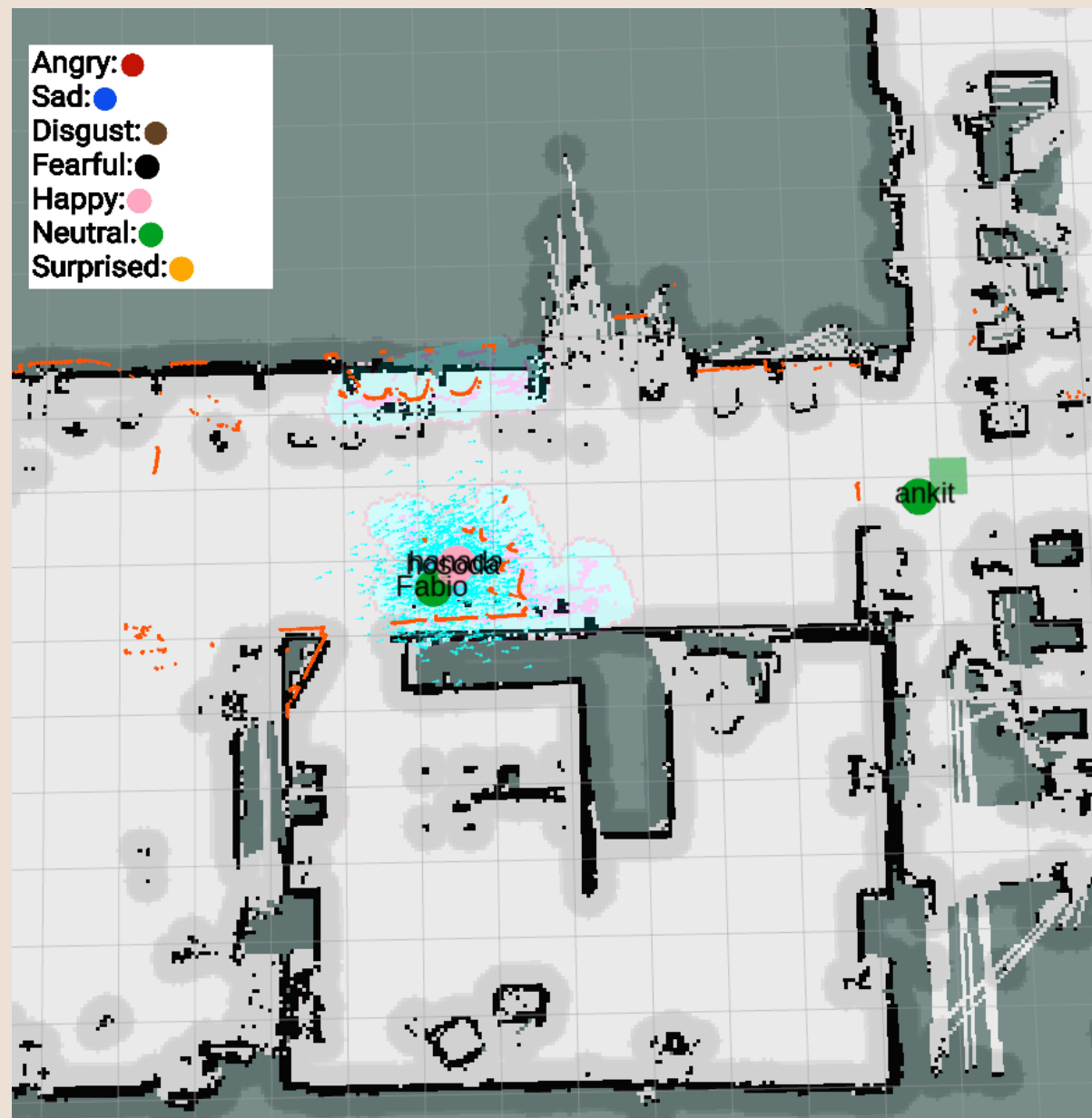
- **mmWave (60 GHz):**

Heart rate (bpm), heartbeat waveform  
proxies, activity state (active/stationary).

- **Use in fusion:**

Health flags and arousal cues (non-  
diagnostic; conservative thresholds).





## Navigation & Spatial Awareness

- **Mapping:**  
Office mapped once with **2D LiDAR** (teleop), producing static map.
- **Localization/Planning:**  
AMCL + move\_base; global + local planning.
- **Obstacle avoidance:**  
Real-time with fused **2D+3D LiDAR** (Teb Local Planner).
- **Markers:**  
People positions and identities overlaid on /map.

# Sensing Fusion (Mood & Health)

- **Inputs:**

FER label, SER label, temperature, heart rate, identity, position.

- **Rule-based logic:**

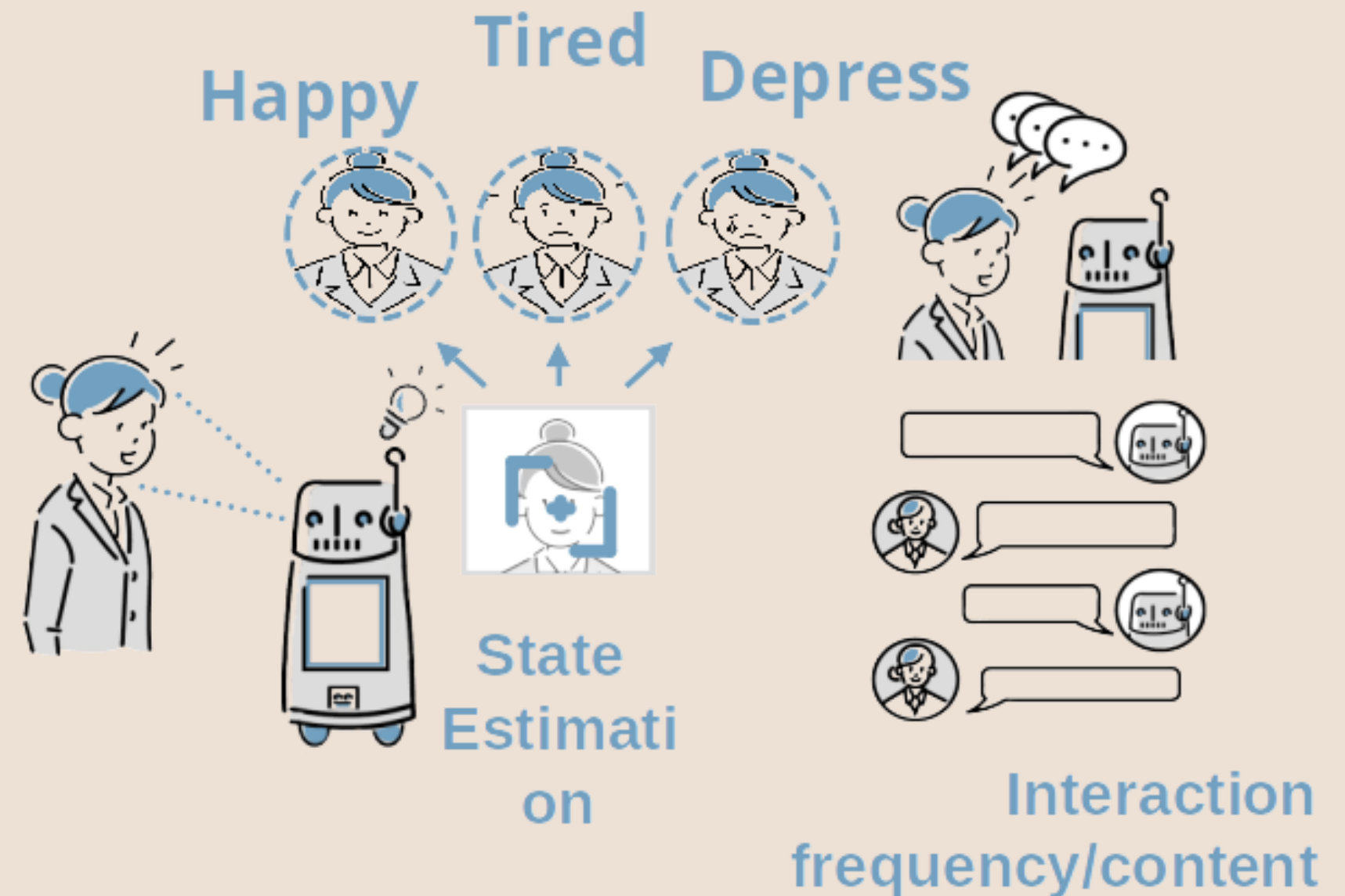
- Same emotion → **intensity = 1.0**
- One neutral → **0.5**
- Opposites (e.g., Happy vs Sad) → **0.25**
- Similar (e.g., Fear vs Sad) → **0.75**

- **Outputs:**

mood (discrete) + intensity, forwarded to interaction.

- **Logging:**

per-user CSV + per-day CSV; RViz markers colored by mood.





# Interaction Management (LLM Policies)

- **Objectives:**

- Improve mood
- Sustain engagement and attention.
- Reinforce positive emotions.
- Encourage health awareness (subtle, non-intrusive).

- **Ethics & Privacy:**

- No storage of raw face/voice/thermal data.
- Only smoothed mood/health features logged.
- No medical claims: robot only suggests, never diagnoses.
- Users may stop interaction anytime ("opt-out" respected).



- **Execution Flow:**

- **Navigation:** Goal sent to *move\_base* with user's 2D position.
- **Visual Check (CLIP):** Ensure user faces robot & is available.
- **Conversation Setup:** Identity & language retrieved → GPT API.
- **Multilingual Prompt + TTS:** Context-based response generated & spoken.
- **User Response:** Speech recorded → Whisper ASR → added to thread.
- **Memory Logging:** Last 3 exchanges stored to re-contextualize future talks.

# Interaction Management (LLM Policies)

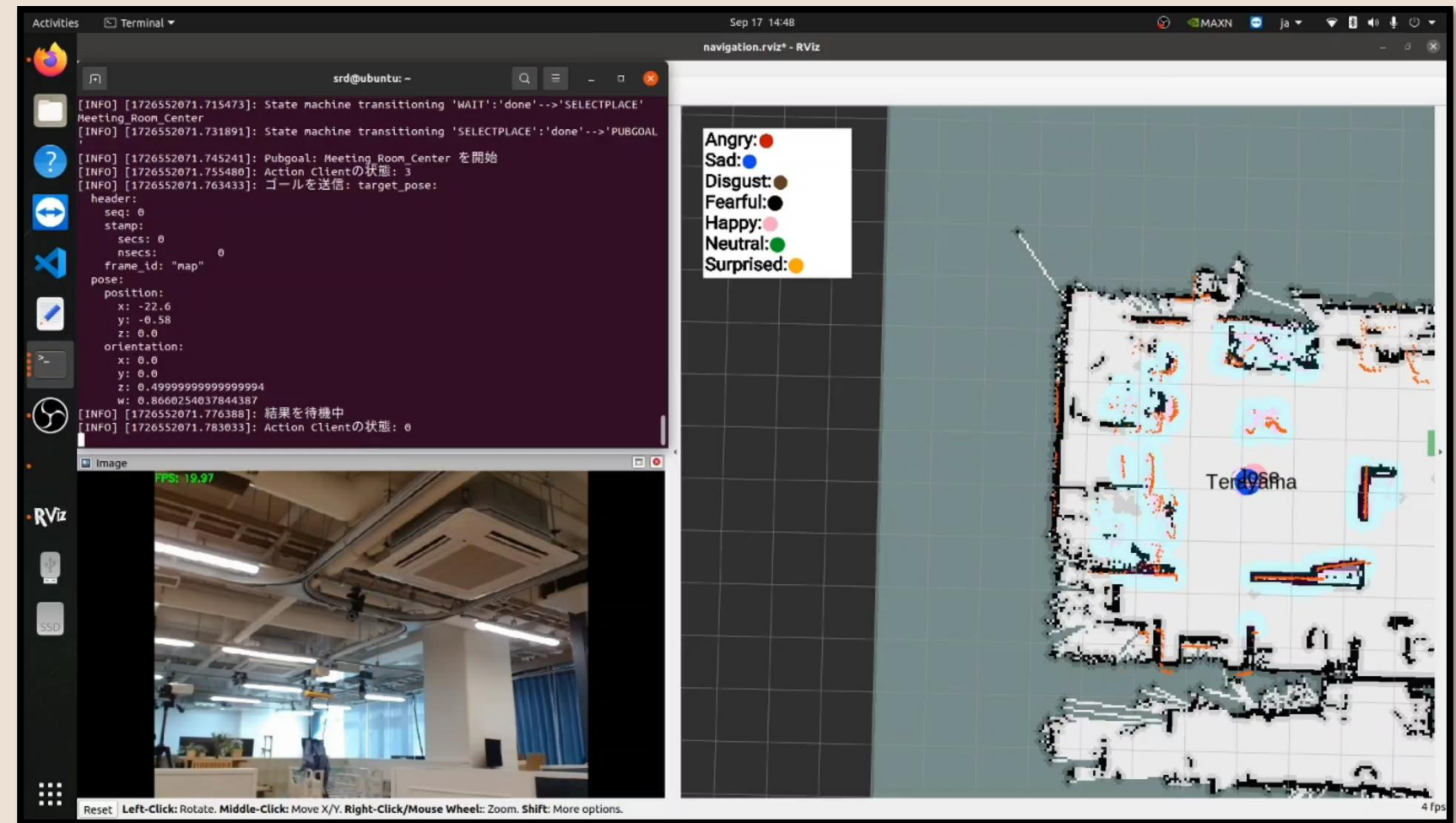
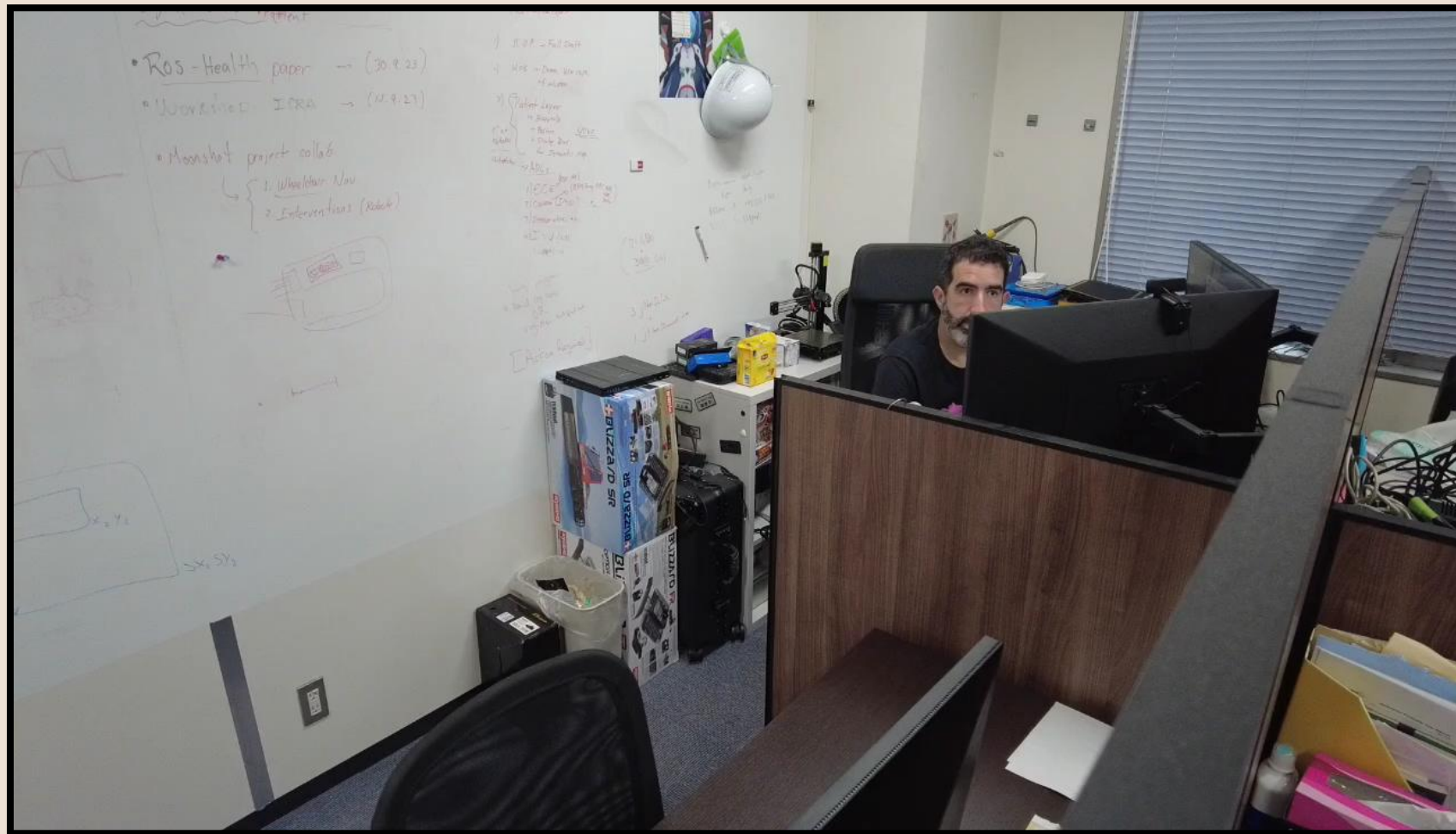


- **LLM-driven dialogues (GPT-based API):**
  - Natural conversations, adaptive style.
  - Cross-talk approach (short turns, no long monologues).
  - CLIP for context aware choice of interaction
- **Two policies tested in experiments:**
  - **No-data policy:**
    - Generic health & well-being support.
    - Topics chosen by robot (e.g., stress, breaks, office life).
  - **With-data policy:**
    - Robot adapts to: [Name, Mood, Temperature].
    - Personalized greetings & language.
    - Mood-sensitive advice (e.g., stress → break, sadness → encouragement).
    - Fever alert if temp > 38°C (non-diagnostic reminder).



# Experiments & Protocols (Overview)

- Experiment 1 — Mood Estimation:
  - 36 recorded interactions; ~20 external observers annotated mood.
  - Compare robot labels vs survey (Accuracy, P/R/F1,  $\kappa$ , CI)



- Experiment 2 — Intervention:
  - Within-subject: **with-data** vs **no-data** (counterbalanced on different days).
  - Post-interaction questionnaires: valence, engagement, usefulness, comfort, acceptance.

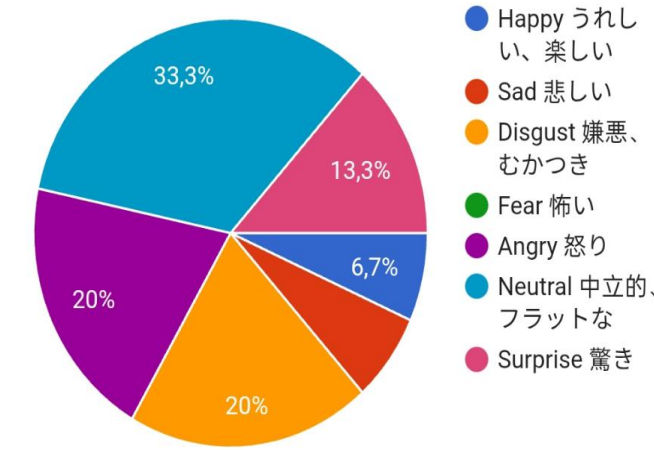
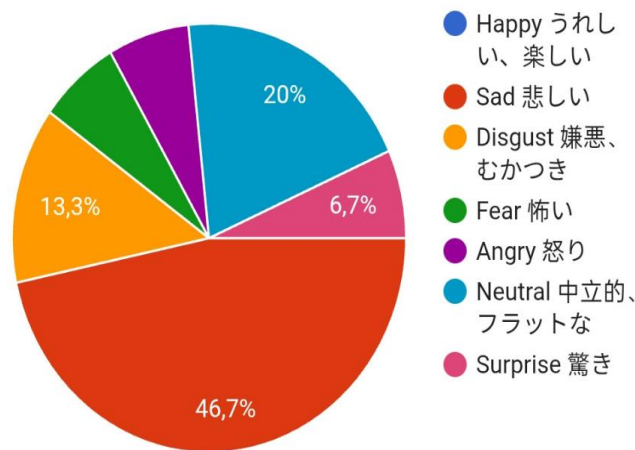


# Results: Experiment 1 (Mood Estimation)

- **6 classes:**
  - Accuracy **58.8%** (95% CI [42.2, 73.6]);  $\kappa = \mathbf{0.42}$  (moderate).
  - Strong on **Happy/Sad**; limited for minority classes.
- **PNN (Positive/Neutral/Negative):**
  - Accuracy **73.5%**,  $\kappa = \mathbf{0.594}$  (moderate–substantial).
  - Clear separation **Positive vs Negative**; **Neutral** remains challenging.
- **Implication:**
  - Use PNN abstraction online; refine discrete mapping with more data.

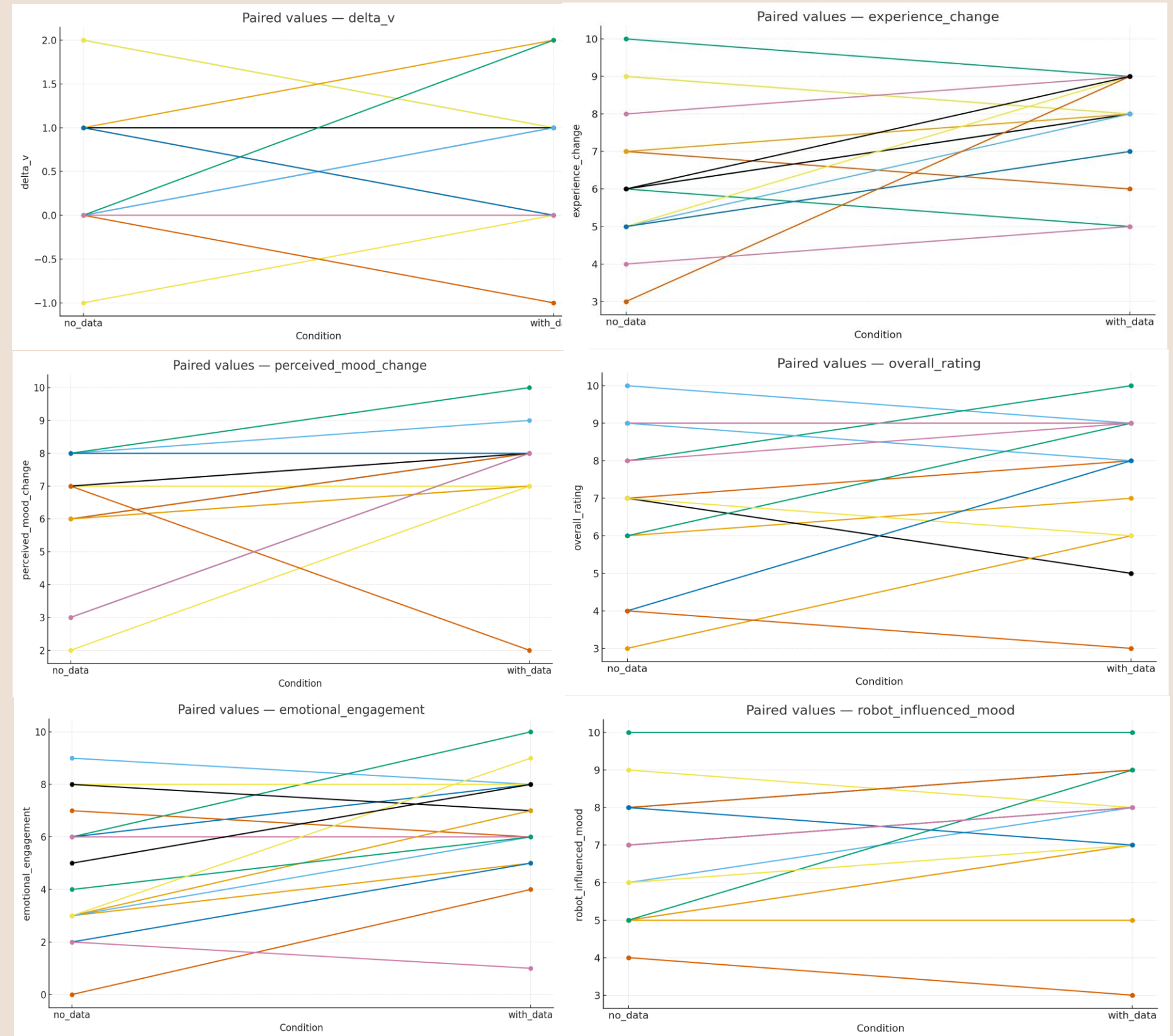
| Class    | Precision | Recall | F1    | Support |
|----------|-----------|--------|-------|---------|
| Angry    | —         | 0.0%   | 0.0%  | 3       |
| Fear     | —         | 0.0%   | 0.0%  | 1       |
| Happy    | 57.9%     | 100.0% | 73.3% | 11      |
| Neutral  | 100.0%    | 30.0%  | 46.2% | 10      |
| Sad      | 50.0%     | 75.0%  | 60.0% | 8       |
| Surprise | —         | 0.0%   | 0.0%  | 1       |

| Class    | Precision | Recall | F1    | Support |
|----------|-----------|--------|-------|---------|
| Positive | 66.7%     | 100.0% | 80.0% | 12      |
| Neutral  | 100.0%    | 30.0%  | 46.2% | 10      |
| Negative | 76.9%     | 83.3%  | 80.0% | 12      |



# Results: Experiment 2 (Intervention)

- **Paired comparison (with-data – no-data):**
  - **Emotional engagement:**  $p = 0.005$ ,  $d_n \approx 0.81$  (large).
  - **Robot influenced my mood:**  $p = 0.027$ ,  $d_n \approx 0.64$  (medium).
  - **Experience change:**  $p = 0.018$ ,  $d_n \approx 0.67$  (medium).
  - **Perceived mood change:**  $p = 0.015$  (Wilcoxon),  $r \approx 0.63$ .
  - Non-sig: overall rating; adaptivity; “understood my emotion” trend ( $p = 0.076$ ).
- **Takeaway:**
  - **Data-adaptive policy** improves engagement and perceived impact.



# Discussion vs Research Questions

- **RQ1 — Estimation:**

- Yes. Multimodal, non-invasive fusion achieved **robust mood estimation** (esp. PNN) in office-like scenarios.

- **Limits:**

- Small sample; neutral ambiguity; conservative fusion; limited deployment time.

- **RQ2 — Intervention:**

- Yes. **With-data** policy yields **significant gains** in engagement and perceived mood influence.

- **Next:**

- Larger datasets, better calibration, learned/probabilistic fusion.



# Future Developments

- **Valence–Arousal (Circumplex):**
  - **Studied theoretically**, not yet implemented; planned as continuous affect layer.
- **Confidence gating:**
  - Fallback prompts when signals are low-confidence.
- **Operations:**
  - Auto-docking; semantic waypoints; scaled evaluations & governance.
- **Health cues:**
  - Calibrate thermal thresholds; validate mmWave HR vs contact reference.
- **Fusion beyond rules:**
  - Bayesian / DS / temporal / lightweight neural fusion.
- **Design management:**
  - Gui
  - Body design
  - Miniaturization and hardware optimization

# Conclusions

**SUNNY** demonstrates a practical path to **privacy-preserving HRI** for mood & health awareness in offices.

- **Experiment 1:**  
Reliable PNN estimation; insights for discrete emotion mapping.
- **Experiment 2:**  
**Adaptive** interactions measurably improve user experience.

**This thesis** establishes the **foundation** for continued research and scaling.

# Thank you for listening!

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